GARSEF
GREATER AUSTIN REGIONAL SCIENCE AND ENGINEERING FAIR
A Program of the Austin Science Education Foundation

SENIOR DIVISION RULES QUICK REFERENCE: COMMON PAPERWORK MISTAKES AND FORMS CHECK

RULES QUICK CHECK: COMMON PAPERWORK MISTAKES

The rules explained below are some that come up frequently during regional and state fair review of projects. This is not a complete list of rules, just the ones most commonly missed. Please review all rules prior to checking student paperwork.

1. **Missing required pre-approval signatures**. Signatures must be obtained **BEFORE** the start of experimentation. Typed names will not be accepted as signatures.
    a. ALL Projects:
        i. Adult Sponsor/Teacher
        ii. Student
        iii. Parent/Guardian.
    b. Some Projects:
        i. IRB, SRC, Qualified Scientist, and/or Direct Supervisor
2. **Missing, incomplete, or incorrect paperwork**.
3. **The project must include all required forms with the appropriate signatures needed for the specific project.**
    a. Parents, relatives, the adult sponsor (teacher), qualified scientist, or direct supervisor of the student may not sign as SRC or IRB members to avoid a conflict of interest.
4. **Research Plan** lacks sufficient details and fails to provide thorough information to support the documentation provided. A properly written research plan must include:
    a. a detailed list of materials including amounts. Include where you will purchase any bacteria or other potentially hazardous biological agents. If you will use E. coli k-12, make sure to list that as the material.
    b. a detailed research plan procedure - projects which cannot be assessed because the research plan is not sufficient will fail to qualify.
    c. clear details about what the student will do when working in a research institution or as part of a larger project. The research plan must identify the student and mentor role in the project.
    d. any additional documentation needed to understand the project including any surveys or tests for human subjects, last year's paperwork for continuation studies, etc.
5. **Human Participant Study**:
    a. Students who work at a Regulated Research Institution (RRI) must have Institutional Review Board (IRB) approval from the RRI. The school IRB is not sufficient. A copy of the RRI IRB approval must be uploaded in STEM Wizard.
    b. Students are prohibited from diagnosing medical conditions, performing medical procedures, or administering medication to human subjects. They may collect data as an observer under

direct supervision of a licensed professional at an RRI. The licensed professional must be named in the research plan and approved by the RRI IRB.

c. On-line surveys require participant consent and must include a statement similar to the one provided by ISEF: "*There is always the possibility of tampering from an outside source when using the internet for collecting information. While the confidentiality of your responses will be protected once the data are downloaded from the internet, there is always a possibility of hacking or other security breaches that could threaten the confidentiality of your responses. Please know that you are free to decide not to answer any question.*" Surveys should require participants to click a button or type a response indicating consent after the statement is read.

d. Any human participant studies involving minors must have parental consent.

e. Form 4 should not have "Not Applicable" selected for any studies involving minors. A form 4 indicates that there are human participants, so the IRB needs to select whether or not minors are involved and if written consent is required for participants 18 years and older.

f. The student researcher is a human participant in their project if they are the subject of the experiment or a variable in the experiment. Examples: Student researcher becomes part of the experiment or tests their design - such as cranking a wheel, jumping, running, kicking a ball - or collecting their own identifiable data such as fingerprints.

g. Engineering projects that use human participants, other than the student and a single adult guardian/sponsor, to test or analyze their product require a Form 4 with IRB review. Also, if students test their own engineering product and it requires them to do a physical activity, they need a Form 4 with IRB review.

h. The above exemption for students testing their own engineering product that does not require a Form 4 does not apply when students use themselves as a test subject in an experiment.

Examples:

Project Type	Project Explanation	Human Involvement	Form 4 Requirement
Engineering	Computer software created by student researcher	20 Students tested the software	Required
Engineering	Computer software created by student researcher	Researcher or teacher tested software	Exempt
Engineering	Student created system where video game only works when a bicycle is used	Researcher or teacher tested the system and there is a physical component	Required
Experiment	Blood pressure tested while playing different video games	Researcher	Required
Experiment	Bacteria cultured from hands after using different hand sanitizers	Researcher	Required

6. **Vertebrate Animal Study**:
    a. Studies performed at a Regulated Research Institution (RRI) require review by an Institutional Animal Care and Use Committee (IACUC). The IACUC review must be uploaded to STEM Wizard. The IACUC review must have the student's name on it or include a letter from the IACUC allowing the student to work on the project.
    b. Students are not allowed to cause pain, distress, or the death of animals in their study. An IACUC may allow this type of study, but it will not be allowed at our fair.
    c. A veterinarian must be consulted and certify experiments that involve supplemental nutrition, administration of prescription drugs and/or activities that would not be ordinarily encountered in the animal's daily life.
    d. Tissue analysis projects are identified as vertebrate animal projects. If the student never comes in contact with live animals and only uses vertebrate animal tissues collected by someone else for the study, this is a tissue study (forms 6A and 6B for **Senior Division**)
    e. Vertebrate Animal projects require Form 5A or 5B (RRI) for **Senior Division**
7. **Potentially Hazardous Biological Agents (microbiology):**
    a. **Bacteria, mold, or other microorganisms can NOT be grown at home. This is an automatic disqualification.**
    b. Microorganisms must be grown in the appropriate BSL setting under supervision, following standard laboratory protocols.
    c. Students are only allowed to work with BSL-1 or BSL-2 organisms. BSL-2 projects may not be done in school labs. These projects must be done at a regulated research institution (RRI) and supervised by a qualified scientist (GARSEF rule beginning 2026).
    d. Unknown organisms cultured from the environment can be treated as BSL-1 only if the petri dishes are sealed shut and never reopened prior to proper disposal. Any project where the student cultures unknown bacteria from the environment and then opens the test tube or petri dish to subculture or analyze the sample is a BSL-2 project.
    e. Yeast, when cultured, requires Form 6A for **Senior Division** and must be done in a school lab or RRI.
8. **Risk Assessment Form 3**
    a. Must be completed for projects that involve chemicals, equipment, or other potential hazards.
    b. Must be completed for any project involving human testing of student-designed inventions, prototypes, or consumer products.
    c. Often missing, and often incomplete without description of safety precautions taken.
9. **Regulated Research/Industrial Form (2C)/Mentorship Disclosure**
    a. Form 2C must be filled out and signed by the mentor or the person who approved the student to work in their lab or business such as the person at HEB that gives the person the right to collect samples there or the research lab supervisor at the university where the person worked. It should NOT be filled out by the parent or teacher.

FORMS (Senior Division)

All Senior Division projects require forms 1, 1A, 1B, and 2A*. The list below consists of forms that are required for projects depending on the experimentation being done; this is only a partial list. For a complete list, check the ISEF Rules and Guidelines. All forms should be completed in STEM Wizard. Links are to sample forms.*

1. Hazardous Materials, Activities, Devices: **Risk Assessment Form (3)**
    - **a.** Chemicals, poison, acid, petroleum products, paint, glue, fertilizer etc.
    - **b.** Power tools, knives, sharp or dangerous objects, machinery
    - **c.** Fire, heat, oven, burner, radiation, electricity, flammable materials/substances
    - **d.** Any other potentially dangerous situation or activity
    - **e.** Exempt Micro-Biology, E-coli k12, algae, bio-fuel, bio-fuel cells, etc.
    - f. Projects that involve human participant testing of any project involving student- designed inventions, prototypes or consumer products
2. Human Participant Study: **Human Participant Form (4) (IRB Pre-Approval)**
    - **a.** Surveys, tests, opinion studies, etc.
    - **b.** Tasting, smelling, exercise, etc.
    - **c.** Medical procedures, blood sugar, blood pressure, DNA kit, etc.
    - **d.** Using other people to test a product you created or purchased
    - **e.** Any research involving humans (including yourself) as test subjects
    - **f.** All studies involving minors who are participants require parental written consent.
    - **g.** Studies where data is collected at a facility where people live or attend programming (day care, nursing home, prison, etc.) will require written permission from the facility prior to beginning experimentation
3. Vertebrate Animal Study: **Vertebrate Animal Form (5A) or (5B) (SRC Pre-Approval)**
    - a. Involves any vertebrate animal (dog, cat, fish, chicken, birds, reptiles, etc.)
    - b. Involves any cephalopods (squid, octopus, etc.)
4. Potentially Hazardous Biological Agents (microbiology): **Hazardous Biological Agents Form (6A) (SRC Pre-Approval)**
    - a. Culturing bacteria, mold, or other microorganisms (can't be done at home)
    - b. E-Coli K-12 does NOT NEED this form, but must be done in a lab
    - c. BSL-2 projects can not be done in a school lab. They must be done in a regulated research institution
5. Tissue and Body Fluid Rules: **Biological Agents Form (6A) & Tissue Form (6B) (SRC Pre-Approval)**
    - a. Blood, blood products (from vertebrate animals or humans)
    - b. Saliva/other bodily fluids (from vertebrate animals or humans)
    - c. Cell or Tissue (from vertebrate animals or humans)
6. Research done outside home, school or field or with contribution from a mentor other than the teacher (including virtual assistance by a mentor): **Regulated Research/Industrial Form (2C)**
    - a. Place of business or commercial/private property (stores, churches, offices, facilities, gyms, hair salon etc.)
    - b. Industrial Location (welding shops, machine shop, mechanic shop, etc.)
    - c. School or University other than the one you attend
7. Continuation Project: **Continuation Form (7)**
    - a. Project related to previous year's research

Senior Division - ISEF Forms with Signatures

Use this document to determine who can sign each ISEF Form.

Form Needed	ISEF Form	Section	Needed By	Dated	Signature Needed	Who Can Sign Here	Name	Email Address
X	1		All Students	Prior To Start Date on Form 1A, Section 7	Adult Sponsor (Teacher)	Teacher		
X	1B	1a	All Students	Prior To Start Date on Form 1A, Section 7	Student (For a Team Project, there should be a Form 1B for each student with their parent's signature. Save as 1 pdf before uploading)	Student		
X	1B	1b	All Students	Prior To Start Date on Form 1A, Section 7	Parent or Guardian	Parent or Guardian		
	1B	2a	Required for projects that need prior SRC/IRB approval BEFORE experimentation (humans, vertebrates or potentially hazardous biological agents).	Prior To Start Date on Form 1A, Section 7	SRC or IRB Chair (SRC/IRB Box 2.A)	SRC Chair - A biomedical scientist with an earned graduate degree or IRB Chair - Usually the medical or mental health professional on the IRB. This can not be the teacher or anyone related to the student(s).		

	1B	2b	Required for projects conducted at all Regulated Research Institutions with institutional approvals for research. (humans, vertebrates or potentially hazardous biological agents).	May be AFTER End Date on Form 1A, Section 7	SRC Chair (SRC Box 2.B)	SRC Chair - A biomedical scientist with an earned graduate degree or IRB Chair - Usually the medical or mental health professional on the IRB. This can not be the teacher or anyone related to the student(s).		
	1B	3	Projects advancing to their Regional Science Fair	AFTER End Date on Form 1A, Section 7 and BEFORE Competition Date	Regional SRC Chair - Projects advanced in STEM Wizard will have this approval after ALL forms are approved.			
	1B	3	Projects advancing to TXSEF	AFTER End Date on Form 1A, Section 7 and BEFORE Competition Date	State SRC Chair - Projects advanced in STEM Wizard will have this approval after ALL forms are approved for TXSEF.			
X	2A		All Students	May be AFTER End Date on Form 1A, Section 7 and BEFORE Competition Date	Student	Student		

	2B		May be required for research involving human participants, vertebrate animals, potentially hazardous biological agents, and hazardous substances and devices.	Prior To Start Date on Form 1A, Section 7	Qualified Scientist. This signature is required if the project needs form 2. (Scientist)	Person who earned a professional degree or has extensive experience in the student's area of research and will supervise the student or coordinate with a direct supervisor to serve in that capacity		
	2B		May be required for research involving human participants, vertebrate animals, potentially hazardous biological agents, and hazardous substances and devices. ONLY Need this signature if Qualified Scientist isn't supervising the student.	Prior To Start Date on Form 1A, Section 7	Direct Supervisor signs ONLY if the Qualified Scientist won't be able to directly supervise the student. (Supervisor)	Person familiar with student's project that will directly supervise work		
	2C		Students who work with a mentor (other than teacher) virtually or on site or who work at a research institution, industrial setting, or any place other than home, school, or field (such as a park or open field).	AFTER End Date on Form 1A, Section 7	Mentor, Supervising Adult, or Whoever gave permission for the student to work at that location	Mentor, Supervising Adult, or Whoever gave permission for the student to work at that location. This should not be the teacher or parent (unless it is the parent's lab or business).		

	3		Recommended for all projects. May be required for projects involving Human Participants, Hazardous Chemicals, Materials or Devices or Potentially Hazardous Biological Agents.	Prior To Start Date on Form 1A, Section 7	Direct Supervisor or Qualified Scientist who will supervise the student during experimentation.	Person who will supervise the student during experimentation		
	4		Required for all research involving human participants not at a Regulated Research Institution. If at a Regulated Research Institution, use institutional IRB approval forms for documentation of prior review and approval.	Prior To Start Date on Form 1A, Section 7	Institution Review Board - This is the section with the check boxes where the risk level and written permission is determined.	Any of the 3 members on the IRB. This can not be the teacher or anyone related to the student(s).		
	4		Required for all research involving human participants **not** at a Regulated Research Institution. If at a Regulated Research Institution, use institutional IRB approval forms for documentation of prior review and approval.	Prior To Start Date on Form 1A, Section 7	Medical or Mental Health Professional (a psychologist, medical doctor, licensed social worker, licensed clinical professional counselor, physician's assistant, doctor of pharmacy, or registered nurse) with expertise related to this project.	Medical or Mental Health Professional on IRB. This can not be the teacher or anyone related to the student(s).		

	4		Required for all research involving human participants not at a Regulated Research Institution. If at a Regulated Research Institution, use institutional IRB approval forms for documentation of prior review and approval.	Prior To Start Date on Form 1A, Section 7	Educator (can not be adult sponsor/teacher)	Another teacher. Can not be the adult sponsor.		
	4		Required for all research involving human participants not at a Regulated Research Institution. If at a Regulated Research Institution, use institutional IRB approval forms for documentation of prior review and approval.	Prior To Start Date on Form 1A, Section 7	School Administrator	Preferably the Principal or Assistant Principal		
	5A		Required for all research involving vertebrate animals that is conducted in a school/home/field research site.	Prior To Start Date on Form 1A, Section 7	SRC Chair (SRC Pre-Approval)	A biomedical scientist with an earned graduate degree. This can not be the teacher or anyone related to the student(s).		
	5A		May be required for all research involving vertebrate animals that is conducted in a school/home/field research site.	Prior To Start Date on Form 1A, Section 7	Veterinarian	Veterinarian		

	5A		May be required for all research involving vertebrate animals that is conducted in a school/home/field research site.	Prior To Start Date on Form 1A, Section 7	Direct Supervisor or Qualified Scientist who will supervise the student during experimentation.	Person who will supervise the student during experimentation		
	5B		Required for all research involving vertebrate animals that is conducted at a Regulated Research Institution. You must include IACUC approval form dated before experimentation.	AFTER End Date on Form 1A, Section 7	Qualified Scientist or Principal Investigator	Person who supervised the student during experimentation		
	6A	3	Required for research involving microorganisms, rDNA, fresh/frozen tissue (including primary cell lines, human and other primate established cell lines and tissue cultures), blood, blood products and body fluids.	Prior To Start Date on Form 1A, Section 7	Direct Supervisor or Qualified Scientist who will supervise the student during experimentation. (QS/DS)	Person who will supervise the student during experimentation		
	6B		Required for research involving fresh/frozen tissue (including primary cell lines, human and other primate established cell lines and tissue cultures), blood, blood products and body fluids. If the research involves living organisms	Prior To Start Date on Form 1A, Section 7	Direct Supervisor or Qualified Scientist who will supervise the student during experimentation.	Person who will supervise the student during experimentation		

			please ensure that the proper human or animal forms are completed. All projects using any tissue listed above must also complete Form 6A.					
	7		Required for projects that are a continuation/progression in the same field of study as a previous project. This form must be accompanied by the previous year's abstract and Research Plan/Project Summary.	Prior To Start Date on Form 1A, Section 7	Student	Student		